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AID Semester 2

LAB EXCERCISE 7

1. Find the correlation coefficient for the following data and explain about the strength of correlation.

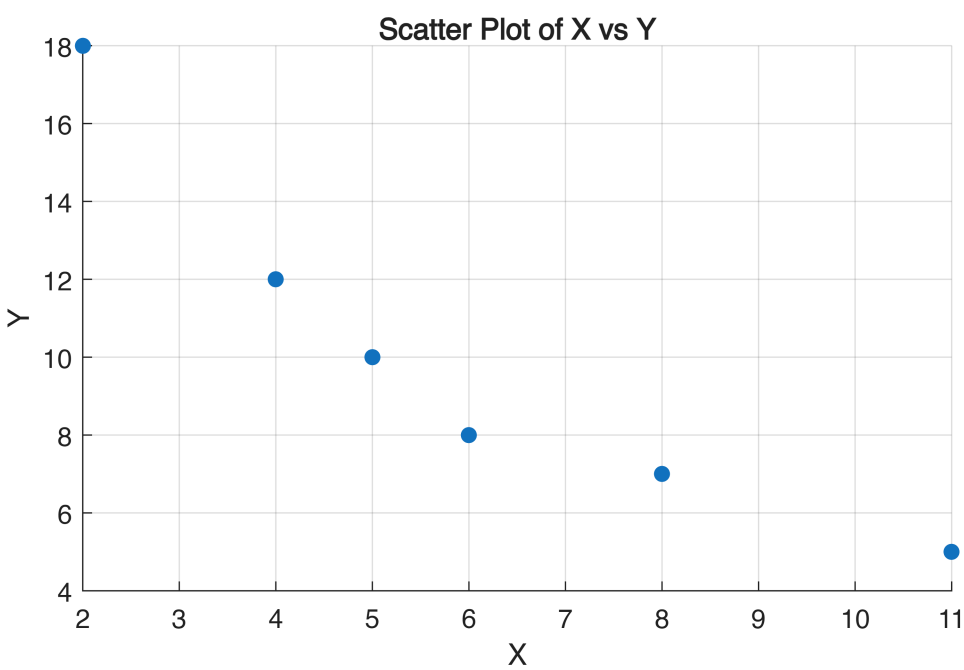
```
X=[2 4 5 6 8 11];
Y=[18 12 10 8 7 5];
C=cov(X,Y)
```

```
C = 2×2
   10.0000  -13.4000
  -13.4000   21.2000
```

```
r=C(1,2)/(sqrt(C(1,1)).*sqrt(C(2,2)))
```

```
r =
-0.9203
```

```
% Plot (optional visualization)
figure;
scatter(X, Y, 'filled');
xlabel('X');
ylabel('Y');
title('Scatter Plot of X vs Y');
grid on;
```




```
if r == 1
    disp('Perfect positive correlation');
```

```
elseif r == -1
    disp('Perfect negative correlation');
elseif r > 0.8
    disp('Strong positive correlation');
elseif r > 0.5
    disp('Moderate positive correlation');
elseif r > 0
    disp('Weak positive correlation');
elseif r > -0.5
    disp('Weak negative correlation');
elseif r > -0.8
    disp('Moderate negative correlation');
else
    disp('Strong negative correlation');
end
```

```
Strong negative correlation
```

2. Janice and Paul did a study on feelings of stress and life satisfaction. Participants completed a measure on how stressed they were feeling (on a 1 to 30 scale) and a measure of how satisfied they felt with their lives (measured on a 1 to 10 scale). The table below indicates the participants' scores. Draw the scatter plot of the data and evaluate the correlation coefficient for the case.

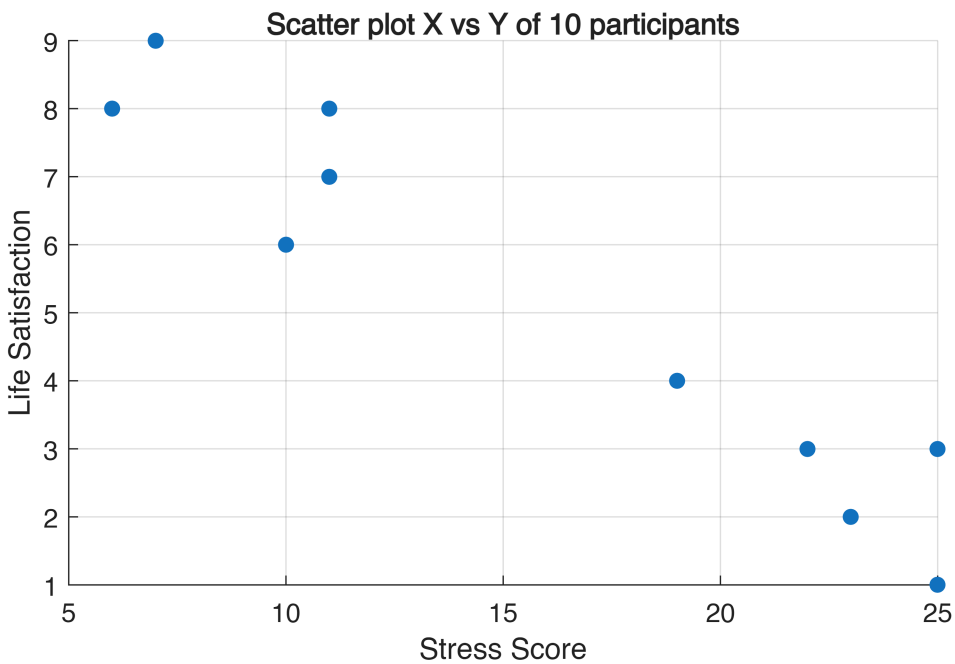
```
X=[11 25 19 7 23 6 11 22 25 10];
Y=[7 1 4 9 2 8 8 3 3 6];
C=cov(X,Y)
```

```
C = 2×2
   58.1000  -20.7667
  -20.7667    8.1000
```

```
r=C(1,2)/(sqrt(C(1,1)).*sqrt(C(2,2)))
```

```
r =
-0.9573
```

```
%plot
figure;
scatter(X,Y,'filled');
xlabel('Stress Score');
ylabel('Life Satisfaction');
title('Scatter plot X vs Y of 10 participants');
grid on;
```




```
if r == 1
    disp('Perfect positive correlation');
elseif r == -1
    disp('Perfect negative correlation');
elseif r > 0.8
    disp('Strong positive correlation');
elseif r > 0.5
    disp('Moderate positive correlation');
elseif r > 0
    disp('Weak positive correlation');
elseif r > -0.5
    disp('Weak negative correlation');
elseif r > -0.8
    disp('Moderate negative correlation');
else
    disp('Strong negative correlation');
end
```

```
Strong negative correlation
```

3. At hogwarts School of Witchcraft and wizardry, students often have a lot of homework. The table below indicates the number of hours students studied and how they perform in their exam in tow of their classes

a. Find the correlations between hours spent studying and how students performed in their portions and defense against the dark arts classes.

```
% Potions data
X1 = [3 15 6 8 4 2 10];
Y1 = [75 95 65 70 85 80 65];
```

```
% Defense Against the Dark Arts data
X2 = [4 12 9 6 2 3 10];
Y2 = [70 98 85 80 65 75 92];

% Correlation calculations
r1 = corrcoef(X1, Y1);
r_potions = r1(1,2);

r2 = corrcoef(X2, Y2);
r_defense = r2(1,2);

fprintf('Potions correlation (r) = %.4f\n', r_potions);
```

```
Potions correlation (r) = 0.2687
```

```
fprintf('Defense correlation (r) = %.4f\n', r_defense);
```

```
Defense correlation (r) = 0.9698
```

b. Which class was more strongly correlated with studying?

```
% Compare strength
if abs(r_potions) > abs(r_defense)
    disp('Potions is more strongly correlated with studying.');
elseif abs(r_potions) < abs(r_defense)
    disp('Defense Against the Dark Arts is more strongly correlated with
studying.');
else
    disp('Both subjects have equal correlation.');
end
```

```
Defense Against the Dark Arts is more strongly correlated with studying.
```

4. The marks of 10 engineering students in Engineering Mechanics(X) and Engineering Mathematics(Y) are given below. Find the correlation coefficient for this data and explain about the strength of correlation.

```
X=[78 40 94 22 76 84 90 62 65 39];
Y=[84 51 91 60 68 62 86 58 53 47];
C=cov(X,Y)
```

```
C = 2×2
  586.2222  284.0000
  284.0000  247.1111
```

```
r=C(1,2)/(sqrt(C(1,1)).*sqrt(C(2,2)))
```

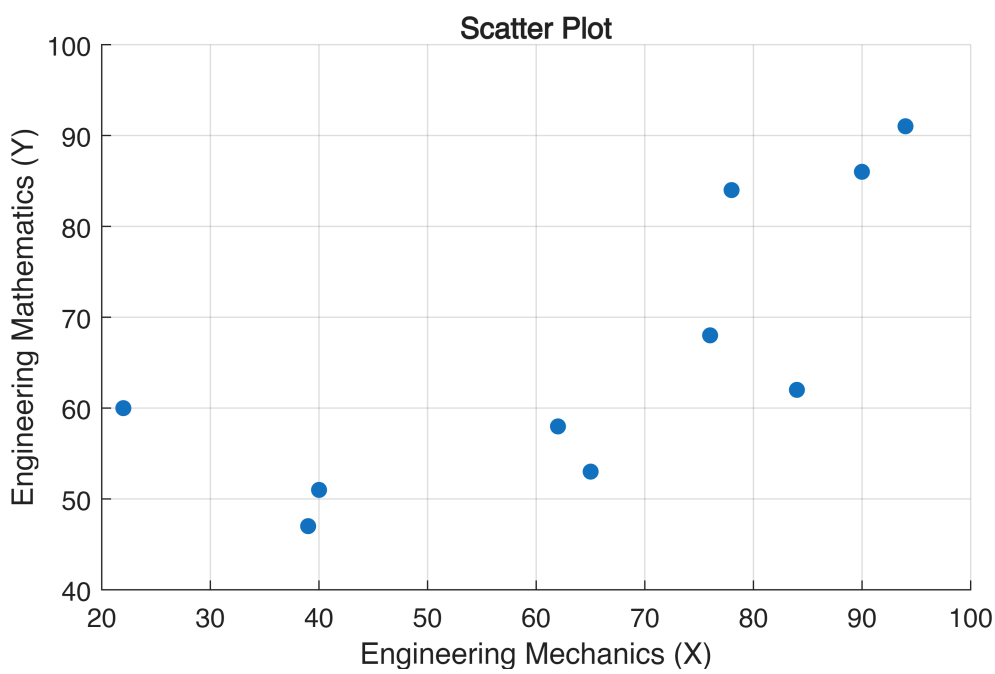
```
r =
0.7462
```

```
% Classification
if r == 1
    disp('Perfect positive correlation');
elseif r == -1
```

```
    disp('Perfect negative correlation');
elseif r > 0.8
    disp('Strong positive correlation');
elseif r > 0.5
    disp('Moderate positive correlation');
elseif r > 0
    disp('Weak positive correlation');
elseif r > -0.5
    disp('Weak negative correlation');
elseif r > -0.8
    disp('Moderate negative correlation');
else
    disp('Strong negative correlation');
end
```

```
Moderate positive correlation
```

```
% Optional plot
figure;
scatter(X, Y, 'filled');
xlabel('Engineering Mechanics (X)');
ylabel('Engineering Mathematics (Y)');
title('Scatter Plot');
grid on;
```



5. Find the correlation coefficient for the following data and explain about the strength of correlation.

```
X=[1 4 7 9 12 15 21 13];
Y=[90 99 85 76 70 65 55 56];
C=cov(X,Y)
```

```
C = 2×2
   40.7857  -92.2857
  -92.2857  255.1429
```

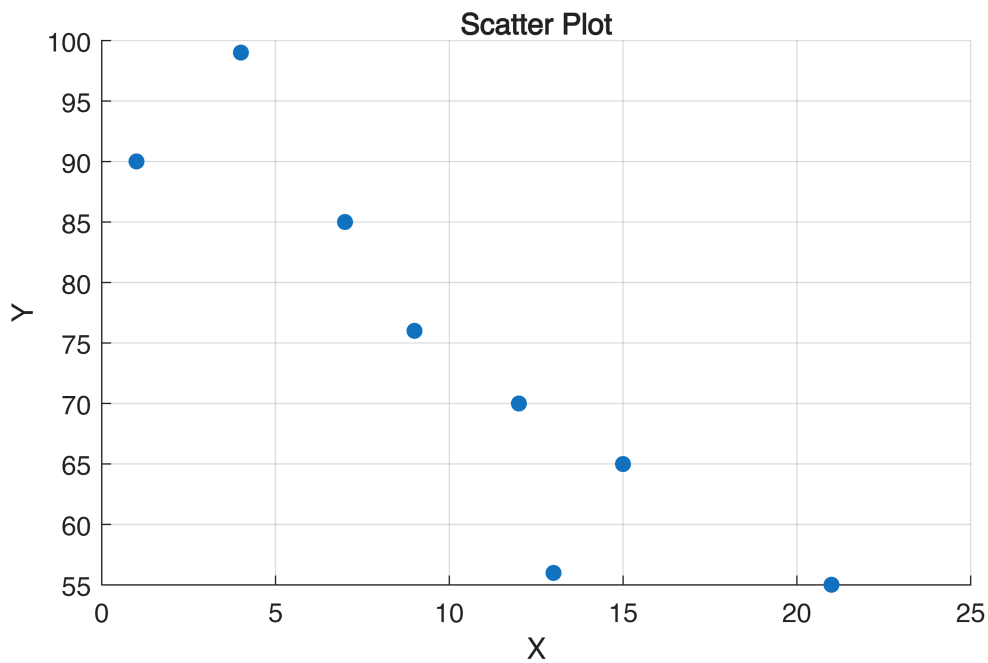
```
r=C(1,2)/(sqrt(C(1,1)).*sqrt(C(2,2)))
```

```
r =
-0.9047
```

```
% Classification
if r == 1
    disp('Perfect positive correlation');
elseif r == -1
    disp('Perfect negative correlation');
elseif r > 0.8
    disp('Strong positive correlation');
elseif r > 0.5
    disp('Moderate positive correlation');
elseif r > 0
    disp('Weak positive correlation');
elseif r > -0.5
    disp('Weak negative correlation');
elseif r > -0.8
    disp('Moderate negative correlation');
else
    disp('Strong negative correlation');
end
```

```
Strong negative correlation
```

```
% Optional plot
figure;
scatter(X, Y, 'filled');
xlabel('X');
ylabel('Y');
title('Scatter Plot');
grid on;
```



6. Find the correlation coefficient for the following data and explain about the strength of correlation.

```
X=[5 8 12 3 7 9 15 1];
Y=[21 28 30 14 23 31 40 9];
C=cov(X,Y)
```

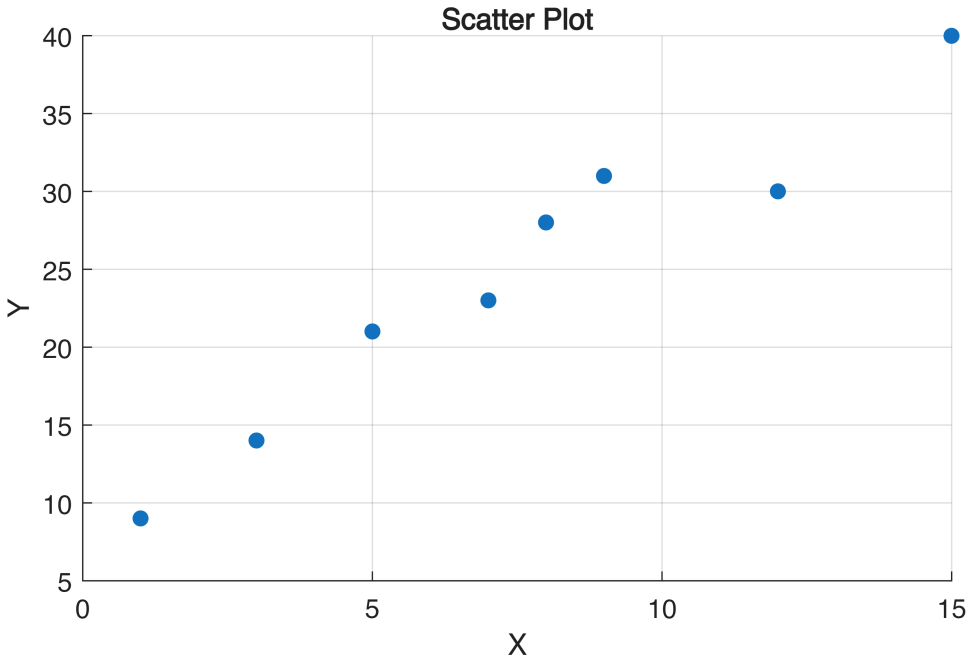
```
C = 2×2
   21.1429   44.2857
   44.2857   98.5714
```

```
r=C(1,2)/(sqrt(C(1,1)).*sqrt(C(2,2)))
```

```
r =
0.9701
```

```
% Classification
if r == 1
    disp('Perfect positive correlation');
elseif r == -1
    disp('Perfect negative correlation');
elseif r > 0.8
    disp('Strong positive correlation');
elseif r > 0.5
    disp('Moderate positive correlation');
elseif r > 0
    disp('Weak positive correlation');
elseif r > -0.5
    disp('Weak negative correlation');
elseif r > -0.8
    disp('Moderate negative correlation');
else
```

```
    disp('Strong negative correlation');
end
```

```
Strong positive correlation
```

```
% Optional plot
figure;
scatter(X, Y, 'filled');
xlabel('X');
ylabel('Y');
title('Scatter Plot');
grid on;
```